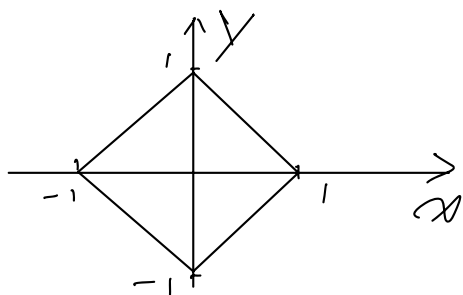


1. l_1 -norm:

The unit ball: $|x| + |y| \leq 1$.



Vertex:
 $(0, 1), (1, 0), (0, -1), (-1, 0)$

Since the matrix is linear transformation, it only alters vertex positions

The new vertex:

$$A(0, 1) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$A(1, 0) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

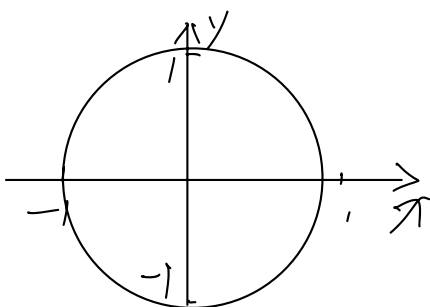
$$A(0, -1) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$$

$$A(-1, 0) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

The induced matrix norm:

$$\|A\|_1 = \max \{ |1| + |0|, |2| + |2| \} = 4$$

l_2 -norm:



The unit ball: $x^2 + y^2 = 1$

Using SVD: $A^T A = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 8 \end{pmatrix}$

$$\det(A^T A - \lambda I) = \lambda^2 - 9\lambda + 4 = 0$$

$$\lambda = \frac{9 \pm \sqrt{65}}{2}$$

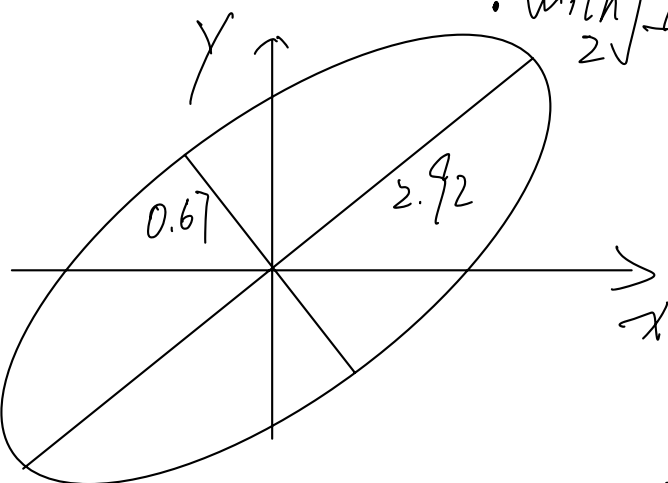
Thus, the ball becomes an ellipse

• with $\frac{\sqrt{9+\sqrt{65}}}{2} \approx 2.92$ as major axis and $\frac{\sqrt{9-\sqrt{65}}}{2} \approx 1.34$ as minor axis

$$\text{Solve } (A^T A - \lambda_1 I)v = 0$$

$$v_1 \approx \begin{pmatrix} 0.25 \\ 0.97 \end{pmatrix} \quad v_2 \approx \begin{pmatrix} -0.97 \\ 0.25 \end{pmatrix}$$

$$u = \frac{Av}{\sqrt{\lambda}} \quad u_1 \approx \begin{pmatrix} 0.75 \\ 0.66 \end{pmatrix} \quad u_2 \approx \begin{pmatrix} -0.66 \\ 0.75 \end{pmatrix}$$



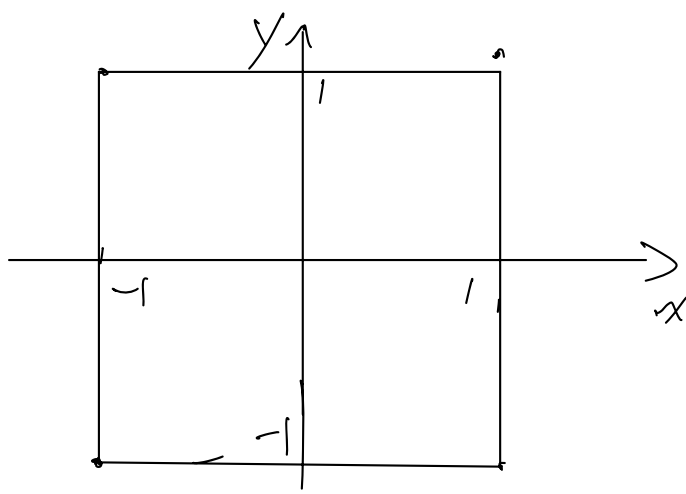
The induced matrix norm:

$$\|A\|_2 = \sqrt{\lambda_1} \approx 2.92$$

Thus, the direction of major axis: $\begin{pmatrix} 0.75 \\ 0.66 \end{pmatrix}$
 the direction of minor axis: $\begin{pmatrix} -0.66 \\ 0.75 \end{pmatrix}$

3rd norm:

The unit ball: $\max\{|x|, |y|\} = 1$



Vertex: $(1, 1), (1, -1), (-1, 1), (-1, -1)$

Since the matrix is linear transformation, it only alters vertex positions

The new vertex:

$$A(1, 1) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

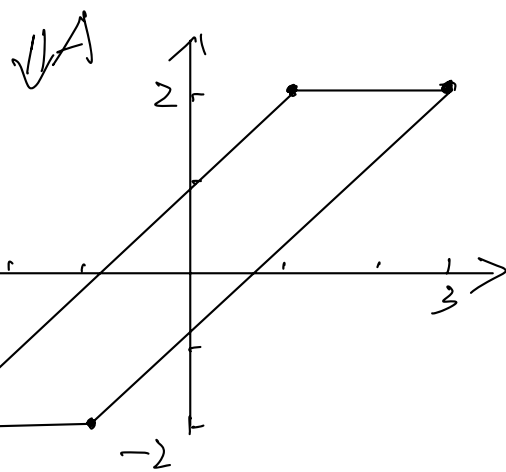
$$A(1, -1) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

$$A(-1, 1) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

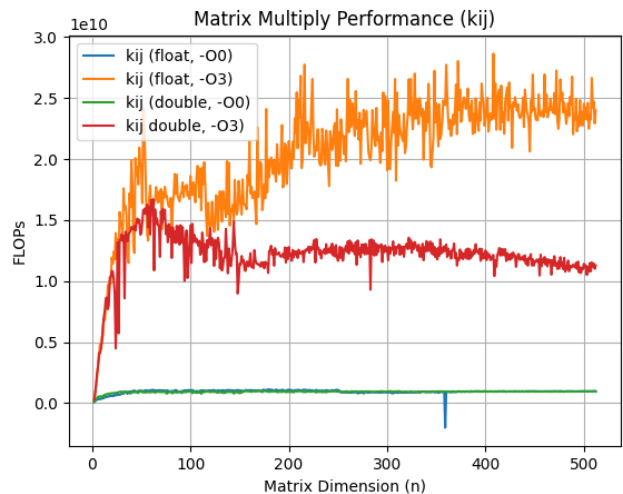
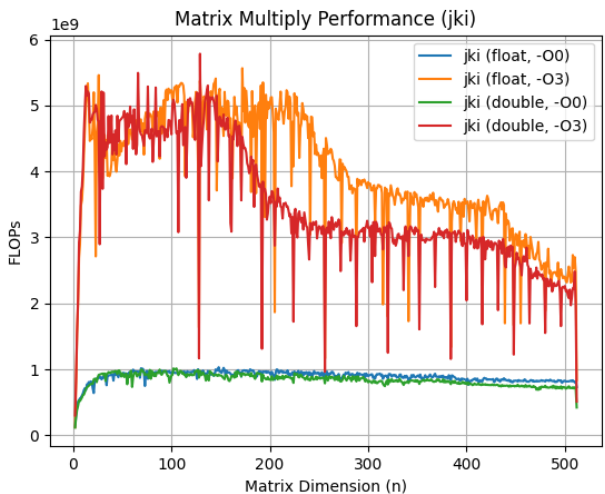
$$A(-1, -1) = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}$$

The induced matrix norm:

$$\|A\|_\infty = \max\{|1|+|2|, |0|+|2|\} = \max\{3, 2\} = 3$$



2.



4. (a) Since (x, y, z) is on the surface $xy + 2xz = 5\sqrt{5} \Rightarrow xy + 2xz - 5\sqrt{5} = 0$

And its distance to the origin is $\sqrt{x^2 + y^2 + z^2}$

We can construct a function

$\mathcal{L}(x, y, z, \lambda) = x^2 + y^2 + z^2 - \lambda(xy + 2xz - 5\sqrt{5})$ which is equal to the square of distance to the origin and look for the minimum

Consider its partial derivative:

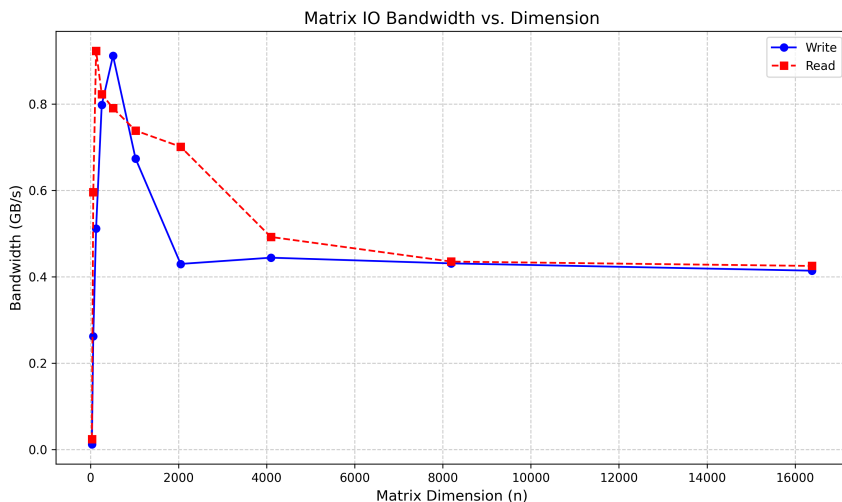
$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda y - 2\lambda z = 0 \\ \frac{\partial \mathcal{L}}{\partial y} = 2y - \lambda x = 0 \\ \frac{\partial \mathcal{L}}{\partial z} = 2z - 2\lambda x = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} = xy + 2xz - 5\sqrt{5} = 0 \end{cases} \Rightarrow \begin{cases} z = \lambda x = 2y \\ 2x - \lambda y - 2\lambda z = 0 \\ xy + 2xz - 5\sqrt{5} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} (2 - \frac{x^2}{2} - 2\lambda^2) = 0 \\ z = \lambda x = 2y \\ xy + 2xz - 5\sqrt{5} = 0 \end{cases} \Rightarrow \begin{cases} \lambda = \pm \frac{\sqrt{2}}{\sqrt{5}} \\ x = \pm \sqrt{5} \\ y = \pm 1 \\ z = \pm 2 \end{cases}$$

Therefore, the coordinate can be $(\sqrt{5}, 1, 2)$ or $(-\sqrt{5}, -1, -2)$

(b) The distance: $\sqrt{x^2 + y^2 + z^2} = \sqrt{(\pm\sqrt{5})^2 + (\pm 1)^2 + (\pm 2)^2} = \sqrt{10}$

5.



6.

File-Based Matrix Swap Performance

